

John Colby

Sr Full Stack Java Engineer

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PROFILE

Results-driven **Senior Full-Stack Engineer** with **13 years** of experience specializing in building **scalable, cloud-native solutions** using **Java** and frontend frameworks like **Angular** and **React**. Proficient in designing and deploying **microservices architectures**, **secure APIs**, and **DevOps pipelines** across **AWS**, **Azure**, and **GCP** platforms. Adept at delivering **compliance-driven applications** for **healthcare** and **financial industries**, with expertise in **HIPAA**, **FISMA**, and **NIST standards**. Skilled in **leading cross-functional teams**, optimizing **system performance**, and leveraging **cutting-edge technologies** to enhance **user experiences**, **streamline workflows**, and ensure **system reliability**. Passionate about **mentoring developers** and driving **innovation** in **high-stakes, data-intensive environments**.

EDUCATION

Bachelor's Degree in Computer Engineering

Aug 2008 – May 2012 | Gainesville, FL, USA

University of Florida

SKILLS

Category	Technologies/Tools
Programming Languages	Java (Spring Boot, Java 11+), Python (3.9), JavaScript (ES6+), TypeScript
Frameworks & Libraries	Spring Boot, Hibernate, Angular (15+), React (17), Vue.js (3), Redux, NgRx, Bootstrap
Cloud Platforms	AWS (S3, RDS, Lambda, ECS, DynamoDB, CloudFormation), GCP (Dataflow, Cloud SQL), Azure (AKS, App Service)
Microservices & APIs	Microservices Architecture, RESTful APIs, Event-Driven Architecture (RabbitMQ, Kafka), gRPC
Containerization & DevOps	Docker, Kubernetes, Helm, Terraform, Jenkins, GitLab CI/CD, AWS CodePipeline
Data Management	PostgreSQL, MongoDB, Redis, Neo4j, MySQL, AWS Aurora
Testing & Quality Assurance	JUnit, TestNG, Mockito, Cypress, Selenium, Unit & Integration Testing
Security & Compliance	OAuth 2.0, JWT, AES-256, SSL/TLS, RBAC, HIPAA Compliance, FISMA, NIST Standards

PROFESSIONAL EXPERIENCE

Senior / Lead Software Engineer

Conduent

Jan 2017 – Present | Austin, TX (Remote)

Project: Government-Backed Healthcare Solutions (Centers for Medicare & Medicaid)

- **Developed a Scalable Patient Management Platform:** Spearheaded the development of a cloud-native patient management system for CMS using Java (Spring Boot), Kubernetes, and AWS. Integrated OAuth 2.0 and JWT authentication, ensuring HIPAA compliance and significantly improving response times for critical patient services.
- **Enhanced Provider Workflow with Seamless Integration:** Partnered with CMS to design a unified interface for EHR, billing, and telehealth services, leveraging Spring Boot and AWS Lambda. Streamlined provider operations, reducing administrative tasks.
- **Implemented Advanced Telemedicine Features:** Engineered secure, real-time telemedicine consultations using Twilio and WebRTC, enabling HIPAA-compliant video calls. Optimized API design to support 160K+ monthly sessions, increasing telehealth adoption by 35% across healthcare providers.
- **Enabled FISMA and NIST 800-53 Compliance:** Directed the implementation of security controls based on federal guidelines, conducting regular risk assessments and monitoring. This effort achieved full compliance with federal policies.

Cloud-Native Microservices and Infrastructure

- **Developed Cloud Infrastructure with Terraform:** Automated cloud resource provisioning using Terraform across AWS environments. Achieved consistent deployments with reduced configuration errors.
- **Designed Healthcare Microservices Architecture:** Designed and deployed 13+ modular microservices using Spring Boot and RabbitMQ, achieving 99.99% uptime and handling 400K+ concurrent healthcare transactions daily. Reduced system downtime by implementing circuit breakers and auto-scaling via Kubernetes HPA.
- **Deployed Serverless Applications:** Leveraged AWS Lambda and ECS to build cost-efficient, scalable serverless infrastructures. Designed Jenkins-based CI/CD pipelines with AWS CodePipeline, reducing deployment time from 45 minutes to under 10 minutes and eliminating manual intervention in production releases.
- **Optimized Database Performance:** Managed healthcare data with PostgreSQL and Redis caching, improving query performance during peak loads.

Data Analytics and Security

- **Developed a Real-Time Analytics Platform:** Engineered a healthcare analytics system using Java, Spring Boot, and AWS Glue. Provided actionable clinical insights, improving patient outcomes.
- **Integrated Role-Based Access Control (RBAC):** Architected fine-grained RBAC policies using AWS IAM, AWS Secrets Manager, and Spring Security, enforcing least privilege access across 1M+ patient records. Reduced unauthorized access attempts by 60% through automated auditing and SOC 2-compliant security monitoring.
- **Advanced Monitoring and Logging:** Integrated ELK Stack and AWS CloudWatch for real-time monitoring and troubleshooting. Reduced issue resolution time.

Enhanced Frontend and User Experience

- **Built a Responsive Front-End for Clinicians:** Developed a React-based, mobile-first UI integrated with GraphQL and REST APIs, reducing clinician task load. Enhanced accessibility by implementing WCAG 2.1-compliant UI components and optimizing for low-bandwidth telemedicine users.
- **Developed Internal Admin Dashboards Using Angular:** Designed secure, modular Angular-based dashboards for CMS administrators to manage provider access, monitor service usage, and configure patient workflow rules. Used RxJS for reactive state management and Angular Material for consistent UI patterns. Integrated seamlessly with backend microservices and existing AWS-based RBAC infrastructure.
- **Designed Cross-Platform Mobile Applications:** Created mobile apps with Flutter and Kotlin for remote patient monitoring and telemedicine, increasing patient engagement.

Leadership and Mentorship

- **Fostered Knowledge Sharing:** Conducted knowledge-sharing sessions on microservices, AWS, and CI/CD practices, reducing onboarding time.
- **Mentored Junior Developers:** Mentored and led junior engineers through code reviews, technical workshops, and hands-on pair programming, accelerating onboarding and skill development. Worked closely with product managers, designers, and DevOps teams to align technical solutions with business objectives. Improved cross-team efficiency through Agile coaching, Jira automation, and streamlined sprint planning.
- **Streamlined Agile Processes:** Utilized Jira and AWS CodePipeline to manage sprints, boosting team velocity.

Previous Experience

Full Stack Java Engineer/Cloud Engineer

AdventHealth

Dec 2014 – Jan 2017 | Altamonte Springs, FL(Remote)

- **Client Data Management System (CDMS):** Designed and implemented a scalable, cloud-native healthcare claims processing platform using Java (Spring Boot) and Azure services (Service Bus, Event Grid, Functions). Integrated Azure SQL Database for structured data storage, ensuring low-latency transactions across distributed systems.
- **Migrated legacy systems:** Modernized legacy infrastructure by migrating from monolithic to microservices architecture with Neo4j and Vue.js, improving data query performance by 30%. Deployed Azure Kubernetes Service (AKS) to containerize workloads, reducing deployment times and enabling zero-downtime releases.
- **CDMS Admin Gateway:** Developed secure API gateways to facilitate interactions between the CDMS UI and various environments (Dev, Test, Prod), leveraging Azure App Service for reliable hosting and Azure Key Vault for secure storage of sensitive API credentials.
- **CDMS Aggregator:** Built a solution using Azure Functions to consolidate related events and improve data transformation workflows, reducing event handling latency.
- **Created dashboards** with Prometheus and Grafana for real-time healthcare data visualization, using Azure Monitor and Azure Application Insights to track application health and performance.
- **Integrated Oracle Database for Historical Claims Data:** Engineered data pipelines to ingest and normalize historical claims from legacy Oracle DB into the CDMS platform. Utilized Oracle PL/SQL procedures to perform complex data transformations and ensure schema consistency. Enhanced reporting accuracy and unlocked 5+ years of archival data for real-time analytics.

Full Stack Developer

- **Client Portfolio Management System (CPMS):** Developed financial advisory solutions using Java (Spring MVC) and MySQL, enabling efficient client portfolio management and data integrity.
- **Built interactive dashboards** with HTML5, JavaScript, and CSS3, enhancing client portfolio and financial analytics capabilities.
- **Financial Data APIs:** Designed RESTful APIs for seamless integration with external financial tools and internal systems.
- **Secure Transaction Framework:** Optimized backend workflows and reduced latency in financial operations with Spring and Hibernate.
- **Implemented robust security measures**, including JWT authentication and SSL/TLS encryption, for secure financial transactions.
- **Streamlined development processes** through version control with Git and Agile sprint management with Jira.